

FINGEAR AI: AN EXPLAINABLE FINANCIAL DIGITAL TWIN FOR PERSONALIZED FINANCIAL WELLNESS, FORECASTING AND DECISION SUPPORT

PHASE I REPORT

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BONAFIDE CERTIFICATE

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“This project addresses the following Sustainable Development Goal(s): SDG 8: Decent Work and Economic Growth; SDG 9: Industry, Innovation and Infrastructure”

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ABSTRACT

FinGear AI is an explainable personal financial intelligence application designed as a Financial Digital Twin for customized financial wellness, forecasting and decision support. It allows users to securely save financial profiles, add income and expense, attach PDF bills, track transactions and recover recently deleted records. It categorizes users into income bands and categorizes transactions into Needs, Wants and Savings categories. FinGear AI does not apply a fixed 50:30:20 rule to every user. Instead, it uses adaptive budgeting targets based on income level. It assesses the spread of expenditure, cash flow, emergency reserves, debt load, saving behavior, consistency of investment and discipline of payment to produce an interpretable financial health score. The score includes explicit reasons at the component level, risk alerts, and actionable recommendations. The application detects abnormal transactions of expenditure and allows the user to verify normal records or discard irregular records from the model's learning. It also provides a personalized expense and cash-flow forecast, starting with rule-based estimates for new users and transitioning to user-specific ARIMA forecasting as transaction history grows. The Financial Digital Twin brings together profile data, transaction behavior, health indicators, anomaly signals and forecast outcomes to power budgeting, goal setting, AI Copilot guidance, what-if simulations and investment decision support. FinGear AI is built with React, Fast API, PostgreSQL and machine-learning services to provide a secure, transparent and scalable platform for better personal financial decision making.

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
AI	Artificial Intelligence
API	Application Programming Interface
ARIMA	Auto Regressive Integrated Moving Average
DBMS	Database Management System
EMI	Equated Monthly Instalment
JWT	JSON Web Token
ML	Machine Learning
NWS	Needs Wants Savings
PDF	Portable Document Format
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Financial wellness means being in a position where you can meet your present commitments, withstand financial shocks, manage your borrowing and make progress toward your future needs. Income alone doesn't reveal these results. High income earners can have volatile cash flow, excessive lifestyle spending or insufficient emergency reserves and low-income earners can be financially resilient through disciplined spending and saving. Therefore, a useful personal-finance system has to analyze behavior in context rather than imposing a single generic rule.

FINGEAR AI builds this requirement on a foundation of explainable financial intelligence. It records a financial profile, maintains a transaction ledger and translates transaction behavior into understandable indicators. The system is based on the principle that fintech should illuminate why the answer is what it is. It doesn't just give a score with no explanation, but the actual split of needs, wants and savings, the right benchmark for the user's income band, flags risks and suggests a gentle course correction.

1.2 PROBLEM STATEMENT

Current financial management systems are mainly designed to record income, expenses and account balances. These features help users monitor transactions, but they often provide disconnected and reactive insights. While users may know how much they spent in a category, they may not know if their pattern of spending is sustainable, the effect it has on their emergency fund, debt position, future ability to save, or overall financial health. Users also lack a unified and personalized system that links financial-profile information, categorized transactions, financial-health assessment, forecasting, goal planning and future decision analysis. Generic budgeting rules and static reports don't take into account differences in income, lifestyle, debt obligations, dependents and financial priorities. Therefore, users may get advice that doesn't fit their financial situation.

Lack of financial awareness and inconsistent financial discipline can lead to unnecessary expenditure, lack of emergency savings, over-indebtedness and delayed progress toward long-term goals. If alerts and guidance are not timely and clear, users may only become aware of financial problems after they have become serious. They may also be unable to recognize unusual transactions, spending trends or the future consequences of their present behavior.

There is a need for an AI-driven Financial Digital Twin that continuously analyzes a user's financial profile and transaction behavior. Such a system should offer personalized, explainable and proactive guidance through adaptive budgeting, anomaly detection, financial health scoring, expenditure forecasting, goal planning and what-if decision support. Closing this gap will enable users to improve their financial discipline, make sound financial decisions and improve their long term financial health.

1.3 EXISTING SYSTEM

Most of the existing personal-finance applications provide only simple means of recording income, expenses, budgets and account balances. They assist users in tracking past transactions, but usually work as standalone tools for tracking, budgeting, forecasting or investment decisions. Users are therefore required to interpret on their own financial data and it is not easy for them to understand the relationship between spending habits, emergency savings, debt obligations, future goals and overall financial wellbeing.

The WONGA system enhances traditional expense tracking by bill-image processing, SMS transaction extraction, automated category classification, personalized budgeting and event-based expense prediction. It shows strong performance in SMS classification, bill extraction, Naive Bayes expense classification and ARIMA forecasting. But it is mainly focused on automating data collection and expense prediction. It does not give a holistic explainable financial health score, adaptive income-tier budgeting, continuous Digital Twin representation, anomaly feedback, what-if analysis, conversational guidance, or integrated investment decision support.

The existing robo-advisory systems also tend to rely on static onboarding questionnaires and periodic risk profiling. They may recommend portfolios but often do not have live cash flow,

changing financial health, transaction behavior and life events integrated continuously. This reduces personalization and makes recommendations less responsive to a user's current financial reality. Opaque AI models can also erode user trust when a system cannot explain why it produced a score, forecast or recommendation.

1.4 PROPOSED SYSTEM

FINGEAR AI is an explainable Financial Digital Twin that builds a dynamic digital representation of an individual's financial condition. A connected financial intelligence system that combines secure user profile, categorized transaction history, income tier, assets, liabilities, emergency fund, financial objectives, investment preferences, anomaly feedback and forecasting result.

The system first creates a complete financial profile and then categorizes the user according to his/her monthly income. It employs customized Needs-Wants-Savings goals, not a universal 50/30/20 rule. For instance, low-income users get a practical, survival-focused allocation while high-income users are nudged to save more and keep lifestyle inflation in check. The system is constantly comparing actual financial behavior with the relevant target and gives gradual, realistic recommendations.

The transaction and expense tracker is the foundation of FINGEAR AI. Manual addition of income and expense, upload of pdf bills, classification of transactions, monthly income update, deletion of entries with confirmation, restore transactions within sixty days. All the validated records are stored in PostgreSQL and shared with health-score, forecasting, planning and Digital-Twin services.

FINGEAR AI produces an understandable financial health score based on spending sustainability, emergency resilience, debt manageability, future readiness, payment discipline and protection. It provides scores for each component, risk flags, a confidence level, and recommended corrective action. The anomaly detection is used to find unusual transactions and the user can confirm a transaction as normal or remove it from the training of the model.

When the user history is limited, the forecasting engine starts with statistical and rule-based estimates. It trains an individual ARIMA model to predict expenditure cash flow and likely financial pressure as more personal transaction data becomes available. These insights are

harnessed in the Financial Digital Twin to enable adaptive budgets, goal planning, AI Copilot explanations, what-if simulations and goal based investment advisory. We will inform investment support with user risk tolerance, financial health, goals, time horizon and portfolio diversification, while ensuring explainability and clear educational disclaimers.

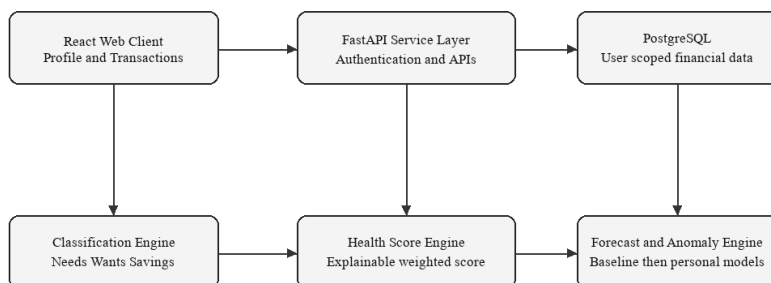


Figure 1.1 FinGear AI basic architecture

1.5 OBJECTIVES

The main goal is to build the financial assessment layer in Phase I of FINGEAR AI. It offers a safe user management system, processes financial profiles with income-tier classification, monitoring transactions and expenses, identifying unusual spending, a clear financial health score and tailored forecasting. Each outcome must be based on data available to the user, traceable to explicit rules or model output, and capable of supporting subsequent Phase II Digital Twin capabilities.

Table 1.1 Phase I objectives

NO.	PHASE I OBJECTIVE
1	Authenticate users and isolate each user's data securely.
2	Capture income, liabilities, assets, emergency fund and financial preferences.
3	Maintain a complete transaction ledger with recovery and bill-reference support.
4	Classify spending into Needs, Wants and Savings/wealth-building categories.
5	Calculate an income-aware and explainable financial health score.
6	Detect anomalous transactions and forecast personal spending behaviour.

1.6 SCOPE

Accounts, profiles, category-based transaction management, document attachments, anomaly feedback, health scoring and forecasting have all been completed in the scope. Health score is for descriptive and educational purposes only and is not investment, credit or legal advice. The persistent store for users, profiles, financial events, transactions, transaction attachments, anomaly feedback, health results and forecast records is PostgreSQL.

Out of scope (deliberately) for Phase II are: Current implementation i.e., adaptive goal and budget planning, AI Copilot interaction, what-if decision simulation, completion of the Financial Digital Twin, and stock or investment advice. The Phase I data model and service boundaries are such that these modules can consume the same validated ledger in future work.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

Uyanahewa et al. proposed a smart personal-finance application named WONGA in [11] that automates expense tracking by bill-image processing, bank-SMS extraction, NLP, machine learning, custom budgeting and calendar-event expense prediction. The system leveraged ANN models for SMS classification, OCR and NER for bill extraction, Multinomial Naive Bayes for expense categorization, ARIMA for short-term expense forecasting, and DistilBERT with regression models for calendar event expense estimation. The study reported 97% SMS binary-classification accuracy, 88% transaction-information extraction accuracy, 98.93% expense-classification accuracy and better ARIMA performance than LSTM for limited expense data. Limitations include reliance on English language documents, Sri Lankan datasets and calendar behavior, and limited support for financial-health evaluation, anomaly feedback, Digital Twin modeling and explainable decision support.

Hyndman and Khandakar [7] proposed the automatic forecasting methods based on the forecast package for R. They automatically select either an exponential-smoothing or an ARIMA model, estimate the parameters, and forecast both seasonal and non-seasonal univariate time series. This study showed that automatic ARIMA and exponential-smoothing procedures are able to generate robust forecasts without the need of a forecasting expert for each series. However, the work is a general forecasting framework and does not address personal-finance context, user-specific transaction classification, financial behavior, goal feasibility or explainable financial recommendations. FINGEAR AI customizes this base with a common forecasting service and per-user ARIMA models for individual expense and cash-flow patterns.

Anshari, Almunawar and Masri [2] studied the use of Digital Twin concepts in robo-advisory systems. Their study suggested that a financial Digital Twin has the ability to combine personal data, real-time information and advanced analytics in order to provide

dynamic and customized financial advice rather than one-off advisory output. The result is a conceptual model showing how Digital Twin-enabled robo-advisors can enhance personalization, client engagement, and financial wellbeing. The main limitation is that the work is exploratory and conceptual, i.e., it does not implement or evaluate a full system for individual transaction, health-scoring, anomaly-detection or forecasting. FINGEAR AI takes this direction further, incorporating the underlying personal-finance data and analytics necessary for a practical Digital Twin.

Bonelli [3] researched Digital Twin personalization and conversational AI in robo-advisory through a scenario-based randomized 2×2 experiment with 336 investment-experienced respondents. The study described Digital Twins as dynamic investor profiles that merge goals, risk tolerance, cash-flow patterns and anticipated life events. The results showed that the personalization of the Digital Twin led to a higher perceived personalization, but also to higher privacy concerns; the conversational AI improved the perceived quality of the interaction, especially in combination with the personalization of the Digital Twin. A major limitation is that the study measured perceptions and behavioral willingness based on scenarios, not actual financial outcomes or real adoption. FINGEAR AI leverages this insight by combining a dynamic profile with explainable conversational assistance, all while maintaining user control and transparent data usage.

Acharya and Bhojak in [1] proposed Financial Doppelgangers framework that integrates Digital Twins, Graph Neural Networks, Distributed Ledger Technology and Explainable AI to detect systemic risk in FinTech ecosystems. The framework works in a continuous feedback loop and was validated over more than 12,000 peer-to-peer lending transactions with macroeconomic and sentiment information. Reported results include an early-warning accuracy of 88%, a 30% reduction in contagion lag, and an AUC of 0.91. Its weakness is that it deals with systemic risk and ecosystem resilience, rather than individual personal-finance management. And the infrastructure is more complex than what's required for a student-oriented personal-finance application. FINGEAR AI applies the principles of continuous feedback and explainability at the level of the individual user.

Jiang et al. [8] proposed an investment-advisory system based on LSTM networks and multi-objective optimization. It uses historical market information, investor behavior, financial

objectives, and risk appetite to create personalized portfolio strategies and dynamically balance risk and return. Simulated results indicated that the system can respond more steadily to market fluctuations than traditional advisory approaches. However, the work is a preprint and is based largely on simulated scenarios – it does not demonstrate real-world suitability, regulatory compliance, explanation quality or integration to a user’s everyday spending and cash-flow condition. FINGEAR AI uses the study as a future investment-advisory reference. It first assesses financial health, emergency reserves, debt safety and user risk profile.

[9] Kong et al. reviewed applications of Large Language Model in finance and investment management. The paper categorizes work on financial text analysis, sentiment analysis, time-series forecasting, financial reasoning, planning, simulations and agent-based modeling. It concludes that LLMs can assist with personalized financial advice, information retrieval, sentiment processing, and code generation for asset-allocation tasks. But this paper also points out that this field of research is still developing and there is no agreement on the most reliable applications. Risks include hallucination, faulty reasoning, data privacy and inappropriate automated recommendations. FINGEAR AI employs an AI Copilot as an explanation and interaction layer on top of structured profile, health and forecast results as opposed to relying on generated text as a standalone source of financial truth.

Das, Markowitz, Scheid and Statman [5] presented a mental-account portfolio framework which combines Mean-Variance Theory with Behavioral Portfolio Theory. The investments are segmented into goal-specific sub-portfolios. The risk is defined as the probability of not achieving the threshold for each financial goal. The study showed a mathematical relation between mental accounts, mean-variance optimization and value-at-risk based risk management . The downside is that it is mainly a theoretical portfolio-allocation framework and does not take into account real time income and expense transactions, changing emergency needs or AI-based financial behavior analysis. FINGEAR AI is able to apply this approach in goal-based investment management, by mapping individual financial objectives, risk and time horizons to a user’s Digital Twin.

Leukam, Koffi and Djagba [10] proposed a goal-based portfolio-optimization method using G-Learning and GIRL, an extension of inverse reinforcement learning. The model

dynamically rebalances portfolio positions to maximize portfolio value by the target date while minimizing periodic contribution requirements. The study reported an improvement in Sharpe Ratio from 0.42 to 0.483 in simulated volatile market. The key limitation is that the results are based on simulation and do not fully account for real transaction costs, market-impact effects, user behavior, regulatory suitability or actual consumer deployment. Once the financial profile and risk constraints are sufficiently established, the method can be used as a future research direction by FINGEAR AI for goal based portfolio decisions.

[6] Feng, Li and Liu proposed a responsible-AI framework for robo-advisory and financial planning. They base their approach on five principles: fiduciary duty, adaptive personalization, technical robustness, ethics and fairness, and auditability. The paper presents a five-level roadmap for AI financial intermediaries, arguing that artificial intelligence can enhance financial access and planning only if design safeguards are in place to prevent information asymmetry, conflicted incentives, and opaque decision-making. The contribution is mainly conceptual and policy-oriented and therefore does not provide results at the implementation level for transaction tracking, health scoring or individual forecasting. FINGEAR AI works on the same principles by way of transparency, adaptive personalization, risk warnings, explanations and user-controlled anomaly feedback.

Černevičienė and Kabašinskas [4] carried out a systematic literature review of 138 studies on Explainable AI in finance. The study identified common financial AI applications such as credit management, stock prediction and fraud detection, and feature importance, SHAP, rule-based methods and combined explanation frameworks are common to improve transparency. The review finds explainability is necessary for trust, accountability and resilient financial decision making. Its limitation is that it synthesizes prior work rather than deploying a new personal-finance system, and it identifies unresolved challenges in measuring explanation quality, fairness and reliability. FINGEAR AI uses this guidance through rule-based health scoring, interpretable risk factors, forecast confidence and visible reasons behind anomaly flags and recommendations.

2.2 COMPARISON AND DISCUSSION

The literature review provides a robust, yet different, foundation for FINGEAR AI. WONGA offers the nearest base for transaction extraction, automated classification, expense prediction and personal budgeting. Hyndman and Khandakar provide the statistical basis for automatic ARIMA forecasting. Digital Twin studies suggest dynamic user representations can improve personalized advisory services, though they also bring up privacy and trust concerns regarding the collection of detailed financial information. The investment and portfolio papers present goal-oriented allocation, risk-return optimization and reinforcement-learning approaches. However, these approaches generally rely on the assumption that the user already has enough investible funds and a stable profile. FINGEAR AI fills this gap by analyzing spending, emergency savings, debt and financial health prior to producing investment recommendations. The LLM and robo-advisory studies show the promise of conversational assistance and AI-driven planning, but also the need for auditability, fairness, user protection and transparent reasoning. The XAI review supports FINGEAR AI's decision to use explainable rules for the core health score and to use ML selectively for anomaly detection, forecasting, AI interaction & future investment optimization.

Existing works typically focus on individual features, such as expense tracking, forecasting, Digital Twin design, investment optimization or explainability. FINGEAR AI integrates these features into a single, user-centric architecture. It contributes to the connection between live transaction behavior, income-tier budgeting, financial-health scoring, forecasting, anomaly feedback, Digital Twin representation, planning and explainable decision support.

2.3 CONCLUSION

The survey of literature confirms the need of unified, explainable and personalized financial-management system. There is a wealth of literature on the value of automated transaction tracking, ARIMA forecasting, Digital Twins, robo-advisory, LLM-based interaction, reinforcement learning and Explainable AI. However, no reviewed system fully integrates adaptive income-tier budgeting, a connected transaction ledger, financial-health scoring, anomaly feedback, per-user forecasting, digital twin modeling, planning, and responsible

financial decision support. FINGEAR AI fills this research gap by providing an explainable Financial Digital Twin that seamlessly links user profile, transactions, health score, anomaly signals, forecast and future decisions forever. The system aims at increasing financial awareness and discipline and at providing transparent, personalized and understandable recommendations.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM LIMITATIONS

FINGEAR AI was created to overcome the shortcomings of traditional personal-finance apps. The common systems provide a transaction ledger, a view of a fixed budget, and summary charts, but don't necessarily combine these functions into a single continually updated financial picture. The user is therefore required to interpret raw transactions, decide whether spending is sustainable and manually compare current behavior with long term goals. The important drawback is the use of the same generic financial rules for people with different incomes and obligations. A fixed allocation such as 50:30:20 may be inappropriate for a user in a survival or baseline income condition, but may be too conservative for a user in a wealth building condition. Current systems also tend to treat mandatory repayments of debt and voluntary savings as the same thing, which can give a misleading impression of financial discipline. Another limitation is the loose coupling of the analytical functions. At the time of budgeting, financial-health scoring and forecasting are implemented as stand-alone features, each using different categories, dates or assumptions. This can lead to values that are not consistent across screens. Traditional trackers also provide limited support for unusual expenses: an abnormal transaction may be flagged as an alert, but the user is not always given a transparent means to confirm it as normal or to exclude it from future learning. Also, today's solutions may depend on incomplete or transient user data. Without persistent, user-scoped historical records, the system cannot develop a reliable spending baseline, measure changes over time or maintain personalized model parameters. FINGEAR AI addresses these limitations with a connected PostgreSQL-backed ledger, adaptive income tiers, explainable scoring, individual category anomaly models and forecast services with statistical and machine-learning methods.

3.2 PROPOSED SYSTEM ARCHITECTURE

The proposed FINGEAR AI architecture is shown in fig. 3.1, which consists of two interconnected sections, namely, Financial Analytics System and Personalized Decision Support. The first part handles financial data and produces analytical results, while the second part combines these results into a unified financial view enabling personalized guidance and decision making. These sections together form the financial digital twin architecture of the application. FastAPI services feed the Financial Analytics System with financial profiles and transaction logs. The data-processing pipeline performs input validation, transaction categorization, feature preparation and organization of the user history. These phases provide a stable financial framework for further analysis. PostgreSQL is the persistent data store, storing the financial records associated with the users, and the historical information needed by the analytical engines.

There are three analytical engines running over this financial information: anomaly detection, forecasting and financial health scoring. The anomaly-detection engine examines spending against the user's behavior history to find transactions that seem out of place. Transaction feedback allows you to acknowledge flagged transactions and also exclude selected records from training the model. The forecasting engine uses baseline statistical methods, ARIMA, and a user-specific Random Forest regressor based on the available financial history. The financial-health engine is powered by explainable, tier-aware weighted rules. The financial-health engine uses the 50:30:20 principle and adaptive allocation targets to measure financial wellness.

The API then connects the two architectural sections through an analytics results flow. The analytical engines generate health scores, prediction and anomaly alerts that help to create the financial state used by the decision-support section. This connection allows personalized views and guidance to be constructed over a shared set of analytical results.

The Personalized Decision Support section is accessed via the React web application. The secure API and JWT authentication are used to route requests to financial information associated with a user. Profile details and income tier classification give context to the

construction of the user's financial state. The Financial Digital Twin is a logical view of shared financial state and services and is not a separate database or independently deployed processing engine and includes transaction and expense tracking and health and forecasting dashboards.

The decision-support functions draw on this shared context to offer goal and budget planning, what-if simulation, AI copilot responses and investment guidance. Goal and budget planning links financial objectives to available resources, and the simulator compares hypothetical changes to the user's baseline financial position. The AI copilot offers tailored responses and explanations, and investment guidance helps with asset-allocation and rebalancing decisions. Thus, the architecture offers a common framework for customized financial wellness and decision support that integrates financial records, analytical results, and user-facing functions.

FINGEAR AI

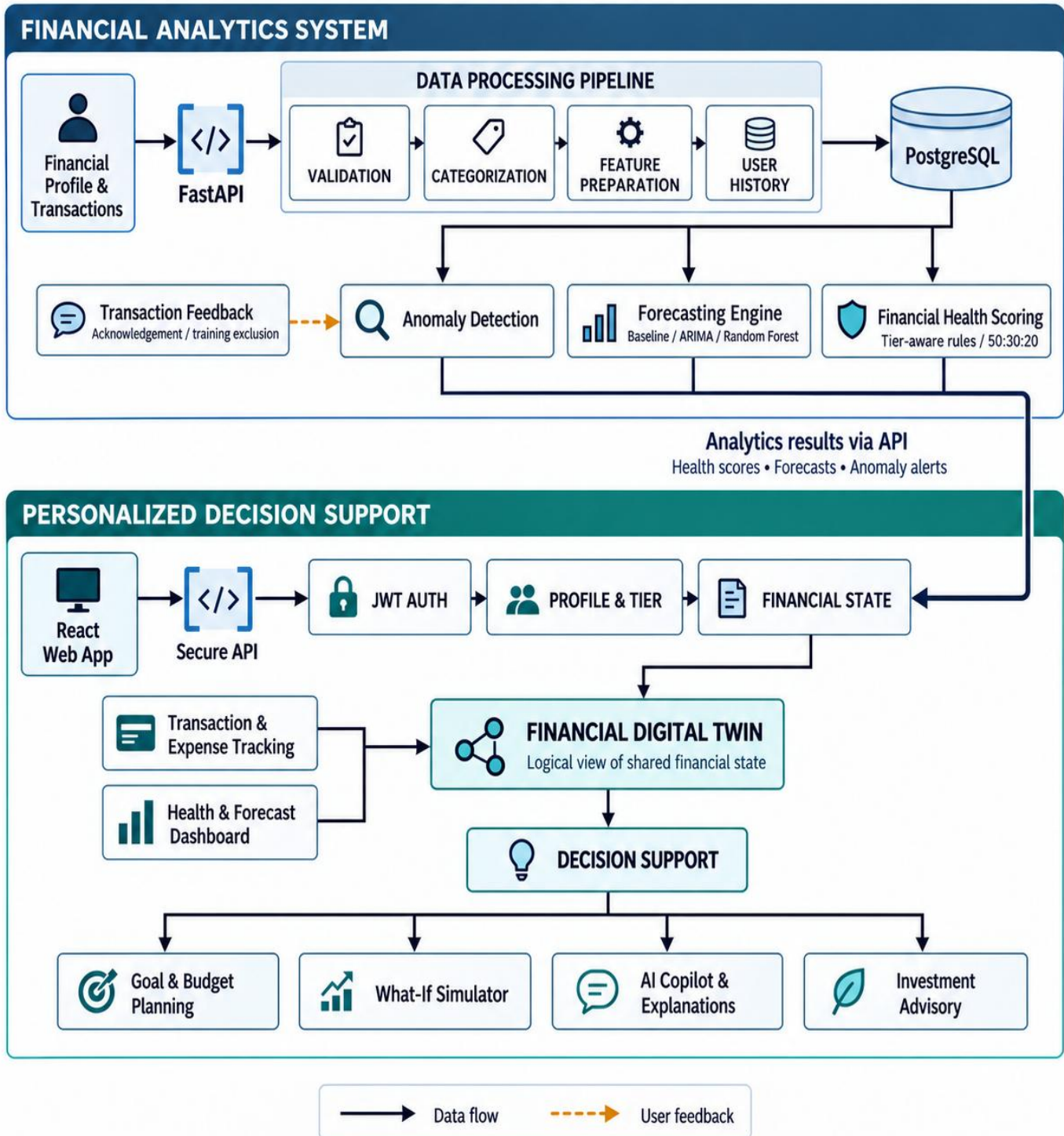


Figure 3.1 FinGear AI system architecture

3.3 SYSTEM REQUIREMENTS

The system shall enable a user to register, authenticate and maintain a private financial profile. It will validate the profile values like income, assets, debts, age, credit score, dependents, salary day, risk appetite, expenses and financial objectives. It shall store all approved information associated with the authenticated user's identity. The system shall provide for creation, viewing, editing, soft deletion and recovery of transactions. Each transaction will include an amount, date, description, type and category and will be available to the health, anomaly and forecasting modules. The system will tag spending into needs, wants and savings or wealth-building categories as identified in the Phase 1 taxonomy. The system shall calculate and describe an adaptive financial-health score and its contributing factors. It will generate expense forecasts from the available historical data, enable the assumed ARIMA and Random Forest capabilities when conditions are met, and present confidence or maturity information where applicable. It will identify anomalies specific to the category and let the user confirm or exclude a transaction from model learning.

3.3.1 SOFTWARE REQUIREMENTS

The software requirements for development and deployment are given below in table 3.1:

Table 3.1 Software Requirements

Category	Requirement
Operating System	Windows or Linux for development; Linux-based containers for deployment
Frontend	React 19, Vite, React Router, Recharts and Lucide React
Backend	Python, FastAPI, Uvicorn and Pydantic
Database	PostgreSQL with SQLAlchemy

Category	Requirement
Authentication	JWT-based authentication and password hashing
Machine Learning	NumPy, pandas, scikit-learn and statsmodels
Development Tools	Visual Studio Code, Git, Docker and Docker Compose
Web Browser	Latest Chrome, Firefox, Edge or Safari with JavaScript enabled
API Communication	REST API with authenticated HTTP requests
Storage	Persistent PostgreSQL storage for profiles, transactions and AI data

3.3.2 HARDWARE REQUIREMENTS

The system is intended to run on standard personal computer hardware in development. The practical minimum requirements are a dual-core processor, 4GB of RAM, 10GB of available storage, and a stable network connection. For the recommended development configuration, a quad-core processor, 8 GB or more RAM and SSD storage is preferable, as the backend, PostgreSQL container and frontend development server may run simultaneously.

For deployment, the backend should have enough memory for concurrent API requests and per-user analytical operations. For a first deployment a small server with two or more virtual CPU cores, 8 GB RAM and persistent SSD storage will suffice. The storage needs to be backed up as the database contains the authoritative financial history and model parameters listed in Table 3.2. As the number of users, transaction history, and forecasting workloads increases, the hardware sizing should be increased.

Table 3.2 Hardware Requirements

Category	Minimum Requirement	Recommended Requirement
Processor	Dual-core CPU	Quad-core CPU or better
Memory	4 GB RAM	8 GB RAM or more
Storage	10 GB free space	SSD with 20 GB or more free space
Network	Stable internet connection	High-speed internet connection
Display	1366 × 768 resolution	Full HD display
Server CPU	2 virtual CPU cores	4 or more virtual CPU cores
Server Memory	4 GB RAM	8 GB RAM or more
Database Storage	Persistent disk storage	SSD with automated backups
Client Device	Desktop, laptop or modern tablet	Desktop or laptop with a modern browser
Deployment Support	Docker-compatible system	Linux server or cloud VM with Docker support

3.3.3 DATABASE DESIGN

The database design is user entity centered. A user has one financial profile and can have many transactions, recurring transactions, income adjustments, budgets, goals, debts, investments, anomaly records, forecast histories and AI-weight records. Ownership is enforced at the database and service levels through foreign-key relationships and user identifiers. The transactions relation is the primary operational ledger. It stores transaction

identifier, user identifier, amount, transaction type, category, description, transaction date, needs-wants-savings classification, recurring information, model-training status and soft-deletion information. Financial amounts are stored as fixed-precision numerics, so calculations are not subject to the vagaries of binary floating-point approximations. See table 3.3.

The financial profile relation stores income, assets, liabilities, emergency funds, expenses, dependants, risk preferences and goals. Relationships between health-score and forecast-history maintain analytical results over time. AI-weight records are learned parameters specific to the user and anomaly feedback records is the user's decision whether the detected event should influence future model learning. Frequent dashboard and forecasting queries are supported by database indexes on user identifiers and transaction dates.

Table 3.3 Phase 1 Data Dictionary

ENTITY	KEY DATA
User	User identifier, name, email, password hash, created timestamp
Financial Profile	Income, tier, assets, debts, emergency fund, profile preferences
Transaction	Date, amount, type, category, NWS bucket, income scope, soft-delete status
Attachment	Transaction reference, file metadata, secure storage reference
Anomaly Feedback	Transaction reference, flag, user decision, training eligibility
Health Score	Component scores, final score, grade, risk flags, confidence, created timestamp
Forecast Run	User model state, horizon, forecast values, confidence and error metrics

3.4 SYSTEM DESIGN

The system design is based on the principle of a common source of truth. A request comes in through an authenticated API route, gets validated by a schema, gets resolved to the current user, and gets processed by a domain service using the user's persisted profile and ledger. The calculation result is passed to the frontend in structured data format and, if necessary, saved as history or feedback.

3.4.1 DATA FLOW DIAGRAM

The primary data flow begins when a user enters profile info or initiates a transaction through the React interface. The frontend makes an authenticated call to the FastAPI API layer. The API validates the request and extracts the user identity from the token. The repository retrieves the user's records from PostgreSQL and sends the validated data to the relevant service as shown in Fig. 3.2.

Profile and transaction data flow to income-tier classifier and explainable health model for health analysis. For anomaly detection, a new transaction flows to the user-specific category brain, compares with the learned category baseline, and returns an anomaly score and explanation. Historical transactions are pre-processed and run through statistical, ARIMA and Random Forest components for forecasting. The forecast and confidence information is then returned to the dashboard. User feedback is looped back into the AI training logs, adjusting the user-category model accordingly.

FINGEAR AI — LEVEL 1 DATA FLOW DIAGRAM

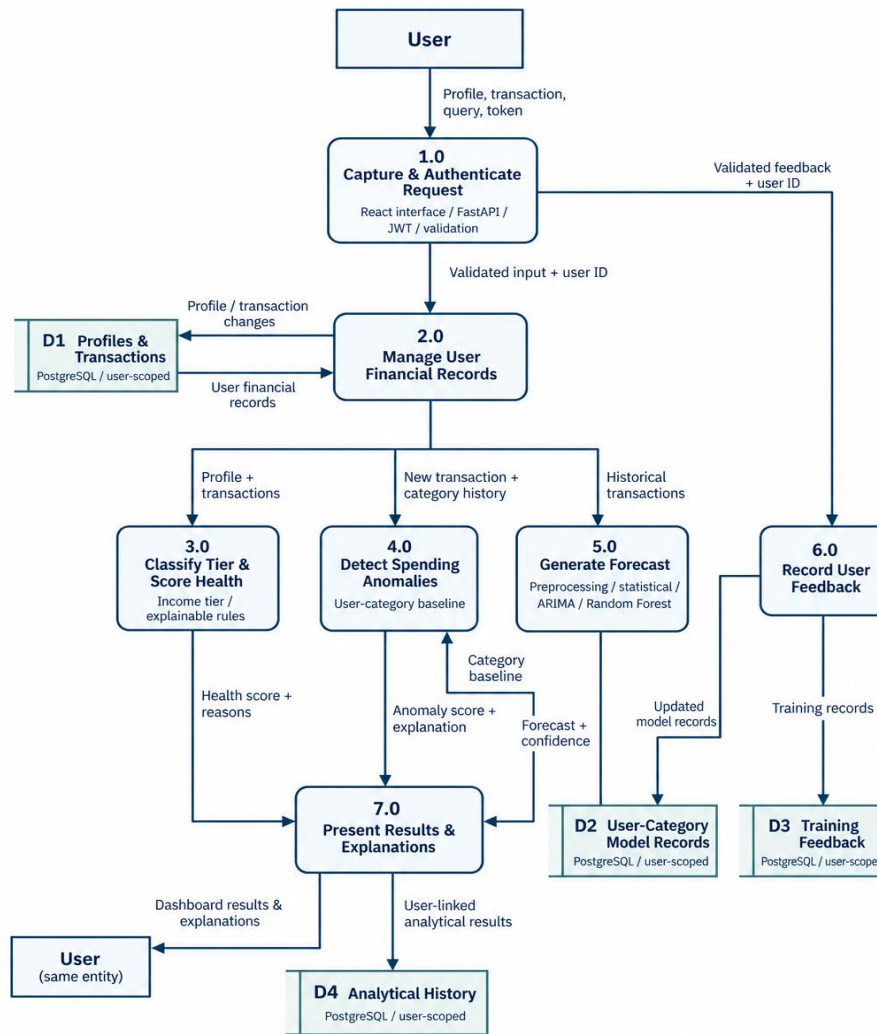


Figure 3.2 FinGear AI Data-flow diagram

3.4.2 ACTIVITY DIAGRAM

The first step of the activity is user authentication. After the user is authenticated successfully, the system retrieves the user profile and financial records. The user can update the profile or add a transaction. The system validates the input, stores the change and initiates the right classification, health, anomaly or forecasting activity. The result is shown with explanatory information.

The user can mark a flagged transaction as normal or exclude it from training for the model. Normal confirmation allows the event to be added to the user's category history. Exclusion keeps the financial record intact but stops it from corrupting the learned baseline. The

activity ends with the presentation of the updated dashboard, score, anomaly state or forecast. Fig. 3.3.

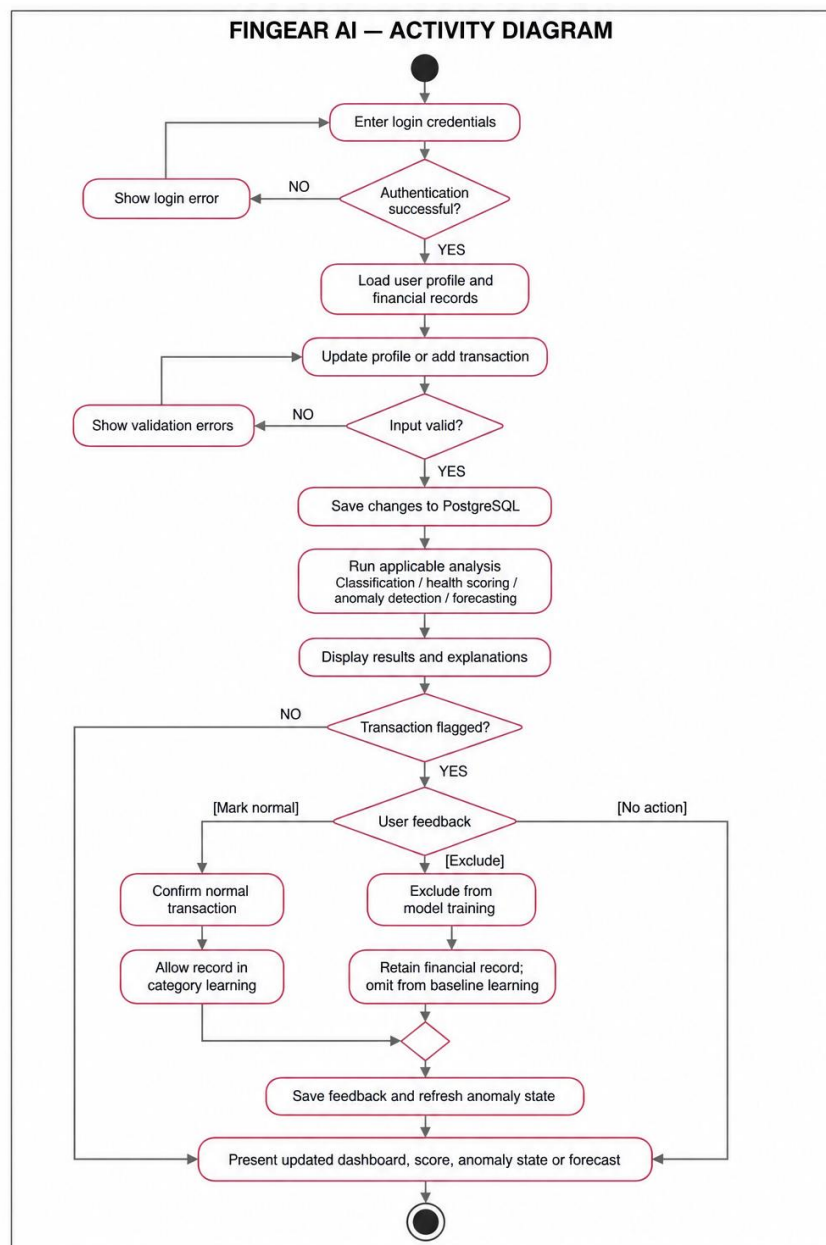


Figure 3.3 FinGear AI Activity diagram

CHAPTER 4

MODULE DESCRIPTION

4.1 USER PROFILE AND FINANCIAL PROFILE MANAGEMENT

This module manages the identity and financial information required for all Phase 1 operations. Users can register and log in securely to create an authenticated session. After authentication, every protected request is linked to a unique user identifier, ensuring that profile information, transactions and analytical results remain private and isolated. The financial profile stores important personal and financial details, including income, occupation, age, number of dependents, assets, liabilities, emergency savings, regular expenses, salary date, financial goals and risk appetite. This information provides the foundation for income classification, financial-health evaluation, anomaly detection and forecasting. Before the profile is stored, all submitted information passes through comprehensive validation. Income, assets, debts, expenses and savings must be valid non-negative monetary values. Age, credit score, number of dependents and salary day are checked against appropriate ranges. Risk-related fields are also validated to ensure that the information is consistent and suitable for financial analysis.

The module distinguishes between declared profile information and actual financial behaviour recorded through transactions. This allows the system to compare the user's stated financial position with observed income, expenses, savings and debt activity. The resulting information is used to determine the user's income slab and configure personalized financial recommendations. PostgreSQL provides persistent storage for all user profiles and related financial records. A user's profile can therefore be accessed consistently across sessions and backend workers. User-scoped database queries ensure that personal information, transaction history, financial-health results, forecasting data and AI parameters remain securely separated between users.

4.2 FINANCIAL HEALTH SCORE

The Financial Health Score module evaluates the user's overall financial condition through an explainable, transaction-aware scoring engine. It produces a score from 0 to 100 and identifies the specific financial factors affecting the result. The engine first determines the user's income tier and then evaluates spending, savings, debt, emergency reserves, planning behaviour and payment discipline according to that tier. Needs–wants–savings budgeting [12], the EMI-to-income (debt-to-income) ratio [13] and emergency reserves sized against essential expenses [12], [14] are established personal-finance principles. The contribution of FinGear AI is not new financial mathematics but their adaptive, explainable integration: targets vary with income tier, and the indicators are combined with safety guardrails into one transaction-driven score.

The five income tiers used by FINGEAR AI are: Survival, below ₹30,000 per month, with a 65:15:20 needs-wants-savings target; Baseline, ₹30,000 to ₹49,999, with a 50:30:20 target; Accumulation, ₹50,000 to ₹79,999, with a 45:25:30 target; Reverse Budget, ₹80,000 to ₹1,24,999, with a 40:20:40 target; and Wealth Building, ₹1,25,000 and above, with a 35:15:50 target as given in table 4.1. Each tier also defines emergency-fund expectations and financial priorities.

Table 4.1 Income-tier allocation targets

TIER	NET MONTHLY INCOME	NEEDS	WANTS	SAVINGS
Survival	Rs. 15,000 - Rs. 30,000	65%	15%	20%
Baseline	Rs. 30,000 - Rs. 50,000	50%	30%	20%
Accumulation	Rs. 50,000 - Rs. 80,000	45%	25%	30%
Reverse Budget	Rs. 80,000 - Rs. 1,25,000	40%	20%	40%

TIER	NET MONTHLY INCOME	NEEDS	WANTS	SAVINGS
Wealth Building	Rs. 1,25,000 and above	35%	15%	50%

The system calculates the average monthly income, needs expenditure, wants expenditure, savings and investments, mandatory EMI, outstanding debt and liquid emergency funds. These values are represented as:

- I = average monthly income
- NE = average monthly needs expenditure
- WA = average monthly wants expenditure
- SV = average monthly savings and wealth-building contributions
- EMI = mandatory monthly debt repayment
- OD = total outstanding debt
- EF = liquid emergency funds

The actual allocation ratios are calculated as:

$$N = \frac{NE}{I}$$

$$W = \frac{WA}{I}$$

$$S = \frac{SV}{I}$$

The income tier provides the target ratios NT , WT and ST for needs, wants and savings respectively. Lifestyle preferences can also adjust the target distribution. A frugal preference reduces the wants target and increases the savings target, while an experience-focused preference increases the wants target while maintaining a minimum savings allocation.

Spending Sustainability

Spending Sustainability contributes 30% to the final score. It evaluates whether the user's needs, wants and savings remain within the recommended income-tier allocation.

The individual allocation scores are calculated as:

$$NeedsFit = \begin{cases} 100, & N \leq NT \\ \min\left(100, \frac{100 \times NT}{N}\right), & N > NT \end{cases}$$

$$WantsFit = \begin{cases} 100, & W \leq WT \\ \min\left(100, \frac{100 \times WT}{W}\right), & W > WT \end{cases}$$

$$SavingsFit = \min\left(100, \frac{100 \times S}{ST}\right)$$

The allocation score is:

$$AllocationFit = 0.25(NeedsFit) + 0.30(WantsFit) + 0.45(SavingsFit)$$

Monthly surplus is calculated as:

$$MonthlySurplus = I - NE - WA$$

The target savings amount is:

$$TargetSurplusSaving = ST \times I$$

The surplus score is:

$$SurplusFit = \min\left(100, \frac{100 \times \max(MonthlySurplus, 0)}{\max(TargetSurplusSaving, 1)}\right)$$

The final Spending Sustainability component is:

$$SpendingSustainability = 0.75(AllocationFit) + 0.25(SurplusFit)$$

This component ensures that the score rewards balanced spending rather than simply measuring income or total savings.

Emergency Resilience

Emergency Resilience contributes 25% to the final score. It measures whether the user's liquid funds can support essential expenses during an unexpected loss of income or emergency.

The essential monthly outflow is:

$$EssentialMonthlyOutflow = \max(NE, 1)$$

The required emergency-fund period depends on income type and dependents:

- Salaried users without dependents: 3 months
- Users with dependents: 4 months
- Users with more than two dependents: 5 months
- Self-employed, freelance or business users: 5 to 6 months

For users in the Survival tier, the emergency reserve must also meet a minimum floor of ₹25,000.

The emergency target is:

$$EmergencyTarget = \begin{cases} \max(25,000, NE \times RiskTargetMonths), & \text{Survival tier} \\ NE \times RiskTargetMonths, & \text{other tiers} \end{cases}$$

Emergency Resilience is calculated as:

$$EmergencyResilience = \min\left(100, \frac{100 \times EF}{\max(EmergencyTarget, 1)}\right)$$

The system also calculates the emergency runway:

$$EmergencyRunwayMonths = \frac{EF}{EssentialMonthlyOutflow}$$

A runway below one month generates a high-resilience-risk warning.

Debt Manageability

Debt Manageability contributes 20% to the final score. It evaluates both monthly debt-service pressure and the total outstanding debt.

The EMI-to-income ratio is:

$$EMIRatio = \frac{EMI}{I}$$

The debt-to-annual-income ratio is:

$$DebtToAnnualIncome = \frac{OD}{12 \times I}$$

The EMI score is calculated using the following relationship:

$$EMIScore = \min\left(100, \max\left(0, \frac{100(0.40 - EMIRatio)}{0.25}\right)\right)$$

This gives the strongest result when EMI is below 15% of income and reaches zero when EMI reaches 40% of income.

The outstanding-debt score is:

$$DebtScore = \min(100, \max(0, 100(1.50 - DebtToAnnualIncome)))$$

The final Debt Manageability component is:

$$\begin{aligned} DebtManageability \\ &= 0.60(EMIScore) + 0.30(DebtScore) \\ &+ 0.10(OnTimePaymentRate \times 100) \end{aligned}$$

A high debt-service ratio or excessive outstanding debt also produces a high-debt-risk warning.

Future Readiness

Future Readiness contributes 15% to the final score. It evaluates savings progress, goal feasibility and investment consistency.

The monthly savings target is:

$$TargetMonthlySaving = ST \times I$$

The savings-target achievement score is:

$$SavingTargetAchievement = \min\left(100, \frac{100 \times SV}{\max(TargetMonthlySaving, 1)}\right)$$

For every active financial goal, the system calculates the required monthly contribution:

$$RequiredMonthlyContribution = \frac{TargetAmount - CurrentAmount}{TargetMonths}$$

If the user's planned monthly contribution meets or exceeds the required amount, the goal receives a feasibility score of 100. Otherwise:

$$GoalProbability = \min \left(100, \max \left(10, \frac{MonthlyContribution}{RequiredMonthlyContribution} \times 100 \right) \right)$$

The average of all active goal probabilities produces the Goal Feasibility score.

Investment consistency is measured across three months:

$$InvestmentConsistency = \min \left(100, 100 \times \frac{ContributionMonths}{3} \right)$$

When active goals are present:

$$\begin{aligned} FutureReadiness \\ &= 0.50(SavingTargetAchievement) + 0.30(GoalFeasibility) \\ &+ 0.20(InvestmentConsistency) \end{aligned}$$

When no active goals are recorded, the goal component is redistributed between savings achievement and investment consistency:

$$\begin{aligned} FutureReadiness \\ &= \frac{0.50}{0.70}(SavingTargetAchievement) + \frac{0.20}{0.70}(InvestmentConsistency) \end{aligned}$$

Payment Discipline and Protection

Payment Discipline and Protection contributes 10% to the final score. It combines payment history, financial planning and protection coverage.

The payment score is:

$$PaymentScore = OnTimePaymentRate \times 100$$

The planning score is based on the number of active financial planning elements:

- Budget, emergency planning and active goals: 100

- Any two active elements: 65
- Zero or one active element: 30

The protection score is 100 when insurance-related expenses are recorded. Provident-fund or fixed-deposit information provides a protection score of 50 when detailed insurance information is unavailable.

The final component is:

PaymentDisciplineProtection

$$= 0.60(\text{PaymentScore}) + 0.25(\text{PlanningScore}) + 0.15(\text{ProtectionScore})$$

Overall Financial Health Score

The raw score is calculated using the five weighted components:

$$\begin{aligned} \text{RawScore} = & 0.30(\text{SpendingSustainability}) + 0.25(\text{EmergencyResilience}) \\ & + 0.20(\text{DebtManageability}) + 0.15(\text{FutureReadiness}) \\ & + 0.10(\text{PaymentDisciplineProtection}) \end{aligned}$$

The final score is rounded and restricted to the range 0 to 100.

The engine applies two financial safety guardrails:

1. If monthly surplus is negative, the final score cannot exceed 59.
2. If EMI is equal to or greater than 40% of monthly income, the final score cannot exceed 49.

These guardrails prevent strong performance in one area from hiding serious affordability or debt problems.

Table 4.2 Financial health score components

COMPONENT	WEIGHT	PRIMARY MEASURES
Spending sustainability	30%	Tier allocation fit and monthly surplus
Emergency resilience	25%	Liquid fund coverage against risk-adjusted essential outflow

COMPONENT	WEIGHT	PRIMARY MEASURES
Debt manageability	20%	EMI-to-income and outstanding debt-to-annual-income
Future readiness	15%	Saving target achievement, goal feasibility and investment consistency
Payment discipline and protection	10%	On-time obligations, planning status and coverage details

Data Confidence

Data confidence is calculated separately from financial health. This ensures that incomplete records reduce the certainty of the result rather than automatically reducing the user's financial-health score.

The confidence formula is:

DataConfidence

$$= 0.40(HistoryCompleteness) + 0.40(CategorisationCompleteness) + 0.20(IncomeVerification)$$

The score is limited to a range of 10 to 100. The system displays:

- High Confidence when the score is at least 80
- Moderate Confidence when the score is between 50 and 79
- Provisional Score when the score is below 50 or the available history is insufficient

The confidence calculation considers the number of tracked days, the percentage of categorized transactions and whether income is supported by transaction records. This allows the user to understand both the health result and the reliability of the information used to produce it.

Adaptive Recommendations

The recommendation engine generates personalized actions from the score components. When wants exceed the income-tier target, the system calculates:

$$ExcessWants = \max(0, WA - WT \times I)$$

The suggested monthly wants reduction is:

$$MonthlyWantsReduction = \min\left(\frac{ExcessWants}{3}, 0.05 \times I\right)$$

This prevents the system from demanding an unrealistic immediate reduction. The recommended amount can instead be redirected towards emergency reserves, savings or systematic investment plans.

The savings gap is:

$$RequiredSavingIncrease = \max(0, TargetMonthlySaving - SV)$$

The emergency-fund gap is:

$$EmergencyGap = \max(0, EmergencyTarget - EF)$$

The system uses these values to generate recommendations for increasing savings, restoring emergency reserves, controlling discretionary spending and improving financial resilience. This makes the Financial Health Score an actionable decision-support module rather than a simple numerical rating.

Financial Grades and Risk Flags

The score is interpreted using four grades as shown in table 4.3:

Table 4.3 Illustrative score interpretation

SCORE	GRADE	SYSTEM INTERPRETATION
80-100	Financially Healthy	Spending, resilience, debt and forward progress are broadly stable.
60-79	Stable, improving	The financial position is generally sound but one or more components need attention.
40-59	Needs Attention	A material cash-flow, savings, debt or planning weakness requires action.
0-39	Financially Vulnerable	The user has immediate resilience or affordability concerns.

The engine also generates risk flags for:

- Emergency reserves covering less than one month of essential expenses
- Missed bills or EMI obligations during the previous 90 days
- EMI pressure of at least 35% of monthly income
- Outstanding debt greater than 1.2 times annual income

4.3 TRANSACTIONS TRACKING

The Transaction Tracking module is the operational core of FINGEAR AI. It records, organizes and processes the user's income and expense activities in a structured financial ledger. Each transaction contains essential information such as transaction type, amount, date, description, category and additional metadata required for financial analysis. Users can record both income and expense transactions. Income Entries can be salary, freelance payments, business income, bonuses, interest and rental income Expense Entries can be housing, groceries, utilities, healthcare, transportation, education, debt repayments, shopping, entertainment and other personal expenses Every transaction is linked to the authenticated user's unique identifier, ensuring that financial records are held private and separate. Before a transaction is stored, the system validates the submitted data. The amount must be a valid positive monetary value, the date must be correctly formatted, the transaction type must be recognized and the category must be suitable for financial classification. Descriptions and optional fields are also processed to ensure that the transaction can be used consistently by the dashboard, financial-health engine, anomaly detector and forecasting service.

Transaction Classification

FINGEAR AI classifies transactions into four mutually exclusive groups as given in table 4.4:

Table 4.4 Transaction Categories

Classification	Representative Categories	Purpose
Income	Salary, freelance income, business income, bonus, interest and rent received	Measures the user's inflow of money
Needs	Housing, rent, groceries, utilities, healthcare, insurance, transport and mandatory EMI	Represents essential or compulsory expenditure
Wants	Dining, shopping, entertainment, travel, hobbies and lifestyle expenses	Represents discretionary spending
Savings and Wealth Building	Savings, investments, SIPs, fixed deposits, mutual funds, stocks, retirement contributions and extra debt repayment	Represents future-oriented financial allocation

Mandatory debt repayment is classified as a Need because it is a compulsory financial obligation. Additional principal repayment beyond the required EMI is classified as Savings and Wealth Building. This distinction prevents the system from incorrectly treating unavoidable debt obligations as voluntary savings.

The category classification is used consistently throughout the application. The same transaction bucket is consumed by the financial-health score, income-tier analysis, dashboard summaries, anomaly detection and forecasting modules. This ensures that all parts of the system use the same interpretation of the user's financial activity.

Transaction Lifecycle

The module supports the complete transaction lifecycle:

1. The user enters the transaction through the frontend.
2. The request is authenticated and linked to the current user.
3. The transaction data is validated by the backend schema.
4. The category is mapped to Income, Needs, Wants or Savings.
5. The transaction is stored in the PostgreSQL database.
6. The transaction is made available to the dashboard and analytical services.
7. The anomaly detector evaluates the transaction against the user's category-specific behaviour.
8. The health-score and forecasting services use the updated ledger in their next calculation.

This is a way to ensure that every transaction becomes part of a unified financial record, rather than being stored separately by individual modules.

Recurring Transactions:

The system supports recurring transactions for regular income and expenses like salary, rent, subscriptions, insurance premiums, loan EMIs, utility bills and investment contributions. Each recurring transaction stores its frequency, amount, category, start date and continuation status. This decreases duplicated data entry and allows predictable transactions to aid cash-flow calculations, health scoring and forecasting.

Income Adjustments:

FINGEAR AI differentiates between temporary and permanent income changes. Bonuses, reimbursements and one-time freelance payments can be applied only to the present month, whereas salary revisions or sustained income changes can roll over into future months. This prevents temporary income from being treated as permanent and enhances cash-flow accuracy.

Bill Attachments and Transaction Metadata:

Users can attach bills or receipts to supported transactions. The system logs related metadata such as the transaction reference, file name, file type, size and upload time. File validation makes sure that only supported files of the right size are accepted, allowing users to hold proof of expenses like rent, medical bills, insurance and loan payments.

Asset-Impact Updates:

Transactions can have an impact on the user's overall financial situation. Investment contributions and savings deposits build assets, while debt repayments lower liabilities. This results in more precise calculations of financial-health and future-readiness.

Soft Deletion and Restoration:

Deleted transactions are soft-deleted instead of being completely removed. They are excluded from active dashboard, health-score, anomaly and forecasting calculations but remain available for restoration during the recovery period. This protects financial history from accidental deletion.

Model-Training Exclusion:

Users can exclude exceptional transactions such as medical expenses, emergency repairs or one-time purchases from AI model training without deleting them from the ledger. The transaction remains visible for record keeping but does not influence learned spending patterns or category baselines.

Integration with Other Modules:

The transaction ledger is shared by the dashboard, income-tier engine, Financial Health Score, anomaly detector and forecasting service. Changes to transactions are reflected consistently across income, expenses, spending classifications, anomaly results and forecasts. This makes the ledger the central source of behavioural evidence for Phase 1.

4.4 ANOMALY DETECTION

The Anomaly Detection module identifies transactions that differ significantly from a user's established spending behaviour. It maintains an independent AI brain for each user and spending category. For example, the Food model learns only Food transactions, while separate models learn Transport, Housing, Utilities, Shopping and Entertainment patterns. Therefore, the same transaction amount may be considered normal in one category but unusual in another.

When sufficient transaction history is unavailable, the system uses a cold-start process based on robust baseline rules and category-level median behaviour. This allows anomaly evaluation to begin even when the user has limited historical data.

As more transactions are recorded, each category model learns the user's typical amount range, frequency and spending distribution. The mature detection process uses a machine-learning ensemble combining category-level clustering and autoencoder-based pattern analysis. The resulting output includes:

- Transaction category
- Anomaly score
- Severity level
- Model maturity status
- Explanation for the detected anomaly

The module functions as a decision-support system rather than automatically rejecting transactions. Users can confirm that a flagged transaction is normal or exclude it from future model training. Exclusion does not delete the transaction from the financial ledger; it only prevents an exceptional expense, such as a medical bill, emergency repair or travel purchase, from influencing the user's normal category baseline.

By isolating anomaly models by both user and category, FINGEAR AI provides personalized anomaly interpretation while preventing one user's spending behaviour from affecting another user's results.

4.5 FINANCIAL FORECASTS

The Financial Forecasting module estimates future expenses, savings capacity and cash-flow pressure using the user's historical financial records. It transforms the transaction ledger into a time-based representation of the user's financial behaviour and generates personalized predictions for daily and monthly planning.

The module uses a progressive forecasting architecture. This allows FINGEAR AI to produce useful results for new users while providing more advanced predictions as sufficient transaction history becomes available.

Forecasting Data Preparation

When the forecast is to be generated, the system retrieves the active transaction history of the authenticated user from PostgreSQL. Deleted transactions and records excluded from model training are removed from the learning dataset. The remaining transactions are organized according to:

- Transaction date
- Transaction type
- Expense category
- Transaction amount
- Day, week and month
- Recurrence and frequency
- Historical category totals
- Recent spending trends
- Income and cash-flow changes

The preprocessing stage converts raw transactions into structured features that can be used by statistical and machine-learning models. This ensures that the forecasting models receive consistent and meaningful financial data.

Progressive Forecasting Stages

The forecasting engine selects a model according to the amount and quality of available history.

1. Statistical Baseline

The statistical baseline is used when the user has limited transaction history. It calculates cautious estimates from recent average expenses, category-level spending and observed transaction frequency. This prevents the system from producing highly complex predictions before enough personal data is available.

The baseline model is useful for:

- Newly registered users
- Users with fewer than seven distinct expense days
- Categories with limited observations
- Periods where transaction history is incomplete
- Initial cash-flow estimation

The statistical baseline provides an early forecast together with a maturity or confidence status so that users understand the reliability of the result.

2. Random Forest Forecasting

Once at least seven distinct expense days are available, the system activates the user-specific Random Forest Regressor. Random Forest is suitable for identifying nonlinear relationships between time, category, transaction frequency and spending amount.

The model can learn patterns such as:

- Higher spending on particular days of the month
- Repeated increases in weekend expenditure
- Category-specific spending behaviour
- Changes in expenses after income adjustments

- Interactions between time periods and spending categories
- Irregular relationships that cannot be represented by a simple average

The Random Forest model is trained using the user's own historical financial behaviour. This allows forecasts to be personalized rather than based on a generic population average.

The forecast transition can be summarized as given in table 4.5 below:

Table 4.5 Forecast Transition based on availability of data

Data Availability	Forecasting Method	Output
Limited history	Statistical baseline	Cautious estimate with provisional maturity
At least seven distinct expense days	User-specific Random Forest	Nonlinear personalized expense forecast
Established multi-month time series	ARIMA-supported forecasting	Trend and time-series projection

3. ARIMA Forecasting

ARIMA is used for established time-series patterns and multi-month projections. It analyzes the sequence of historical expenses to identify:

- Recent trends
- Repeated time-series behaviour
- Changes in expense levels
- Temporal dependency between previous and future values
- Expected movement across future periods

ARIMA is especially useful when the user's expense history contains sufficient chronological data. It complements the Random Forest model by focusing on time-dependent patterns, while Random Forest captures nonlinear relationships among multiple features.

The combined forecasting architecture allows the system to use statistical, machine-learning and time-series techniques according to the structure of the available data.

Daily Forecasts

Daily forecasts estimate the user's expected expense for upcoming days. They support short-term financial planning by showing the likely amount of spending before it occurs.

Daily predictions may consider:

- Average expense for comparable days
- Recent category activity
- Day-of-week behaviour
- Periodic salary or income patterns
- Recurring transactions
- Recent spending increases or reductions
- Historical expense frequency

The daily forecast output includes the forecast date, expected expense amount, model status, confidence level and relevant category or trend information.

This helps users identify periods of increased financial pressure and adjust discretionary spending before the expense occurs.

Monthly and Multi-Month Forecasts

Monthly forecasting aggregates expected daily and category-level expenses to estimate future cash-flow movement. It can project:

- Total expected expenses
- Expected income
- Monthly surplus or deficit
- Estimated savings capacity

- Category-level spending
- Upcoming recurring obligations
- Emergency-fund or debt-payment pressure

The monthly cash-flow relationship is represented as:

$$\begin{aligned} \textit{ProjectedMonthlySurplus} \\ &= \textit{ProjectedIncome} - \textit{ProjectedNeeds} - \textit{ProjectedWants} \\ &\quad - \textit{ProjectedDebtPayments} \end{aligned}$$

The projected savings capacity is calculated from the expected surplus and planned savings contributions:

$$\textit{ProjectedSavingsCapacity} = \max(0, \textit{ProjectedMonthlySurplus})$$

These values help the user understand whether upcoming income is likely to cover essential spending, discretionary expenses and financial commitments.

Category-Level Forecasting

The system can examine expenses by category instead of relying only on one overall total. Category-level forecasting helps identify whether future pressure is likely to come from:

- Housing
- Groceries
- Utilities
- Healthcare
- Transportation
- Debt payments
- Shopping
- Entertainment
- Travel
- Investment and savings contributions

Category-level results allow users to understand not only how much they may spend, but also where the expected spending is likely to occur. This improves the usefulness of recommendations generated by the health-score and budgeting modules.

Forecast Maturity and Confidence

Every forecast includes information about its model maturity and reliability. The confidence status is determined by the amount of available transaction history, the number of distinct expense days, transaction consistency, category completeness and the stability of observed spending patterns.

The system classifies forecasts as provisional, developing or mature. Provisional forecasts are generated when only limited financial data is available. Developing forecasts are based on an expanding personal transaction history, while mature forecasts are supported by established time-series patterns and user-specific machine-learning models. This classification prevents users from interpreting predictions based on limited records with the same certainty as forecasts generated from a long and consistent financial history.

Forecast Updating

Forecasts are recalculated whenever significant financial information changes. New income or expense transactions, modifications to recurring transactions, income adjustments, restored transactions, category changes, model-training exclusions and updates to the financial profile can all trigger a revised forecast.

This allows the forecasting module to respond continuously to the user's current financial condition. For example, a salary adjustment, recurring EMI or unusual expense can influence future projections without requiring the user to manually rebuild the forecast. The updated results are then reflected across the forecast interface and other analytical modules that use the shared forecasting service.

Integration with Other Modules

The forecasting service shares the same validated ledger used by the dashboard, financial-health engine and anomaly detector. This ensures that all modules work with consistent income, expense and category data.

The forecast interacts with the other Phase 1 modules in the following ways:

- The profile module provides income, debt, savings and financial-context information.
- The transaction module provides historical income and expense records.
- The recurring-transaction module provides predictable future commitments.
- The anomaly module identifies unusual transactions that may require special handling.
- The health-score module uses projected cash flow to support future-readiness analysis.
- The dashboard presents daily and monthly forecast results.
- The financial-planning interface uses forecasts to identify expected surplus or financial pressure.

Transactions excluded from model training remain part of the user's financial record but do not distort the learned spending baseline. This allows the forecasting service to distinguish ordinary spending behaviour from exceptional events.

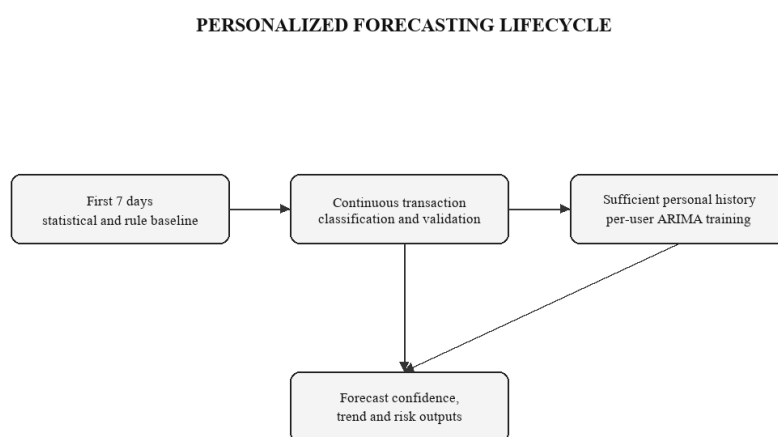


Figure 4.1 Personalized forecasting lifecycle

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSIONS

5.1 IMPLEMENTATION RESULTS

The implementation was evaluated across three principal modules: financial health scoring, transaction anomaly detection and expense forecasting. The validation assessed rule alignment, classification performance and forecasting error using module-specific benchmarks. The following results summarize the supplied validation report and accompanying visualizations.

5.1.1 IMPLEMENTATION OF FINANCIAL HEALTH SCORING

The financial-health engine was evaluated across five income tiers to examine whether its adaptive scoring rules produced the expected outcomes for the tested financial profiles. The validation reported 100% five-tier adaptive alignment, indicating that all evaluated cases satisfied the specified tier-based expectations.

Table 5.1 Validation of users across different categories

Income tier	Monthly income	Health score	Reported assessment
Survival	₹22,000	38/100	Financially Vulnerable
Baseline	₹42,000	72/100	Stable, Improving
Accumulation	₹72,000	92/100	Financially Healthy
Reverse Budget	₹1,15,000	97/100	Financially Healthy
Wealth Building	₹2,40,000	97/100	Financially Healthy

These results demonstrate the operation of the scoring engine across different financial profiles and income classifications. The scores reflect the complete inputs supplied for each

test case; they should not be interpreted as scores determined by income alone. The tier mechanism adjusts the evaluation benchmarks, while the user's financial indicators determine the resulting assessment. The critical-debt guardrail also achieved a 100% trigger rate in the tested cases. Under the tested condition of a debt-to-income ratio exceeding 40%, the engine capped the health score at 49 or below. This verifies that the implemented safeguard prevents a critical debt condition from being concealed by stronger performance in other scoring components. The reported alignment measures consistency with the defined rules, rather than independent validation of financial wellness against an external standard.

5.1.2 IMPLEMENTATION OF TRANSACTION ANOMALY DETECTION

The anomaly-detection ensemble was evaluated on 300 labeled transactions comprising 240 normal transactions and 60 anomalies. The validation report identifies the evaluated approach as a K-Means and neural-autoencoder ensemble. It achieved **95.67%** classification accuracy, correctly classifying 287 of the 300 transactions. The confusion matrix shows that all 240 normal transactions were correctly classified, while 47 of the 60 anomalies were detected. There were no false positives and 13 false negatives. Consequently, precision reached 100%, recall reached 78.33%, and the F1-score was 0.8785. These results indicate that every flagged transaction in this evaluation was an actual anomaly, although some anomalous transactions remained undetected. The score-distribution visualization shows substantial separation between normal and anomalous transactions at the displayed decision threshold of 0.65. The reported ROC-AUC of 0.9992 indicates strong ranking separation across thresholds. However, recall at the selected threshold confirms that high separability does not automatically result in the detection of every anomaly. The transaction scatter plot illustrates the distribution of flagged spending events. Its 253 unflagged transactions comprise 240 true normal transactions and 13 missed anomalies; therefore, the plotted label "Normal Transactions" should be understood as the model's predicted classification. Overall, the results demonstrate strong precision and an absence of false alarms on the evaluated dataset, with improved sensitivity to missed anomalies remaining an area for refinement.

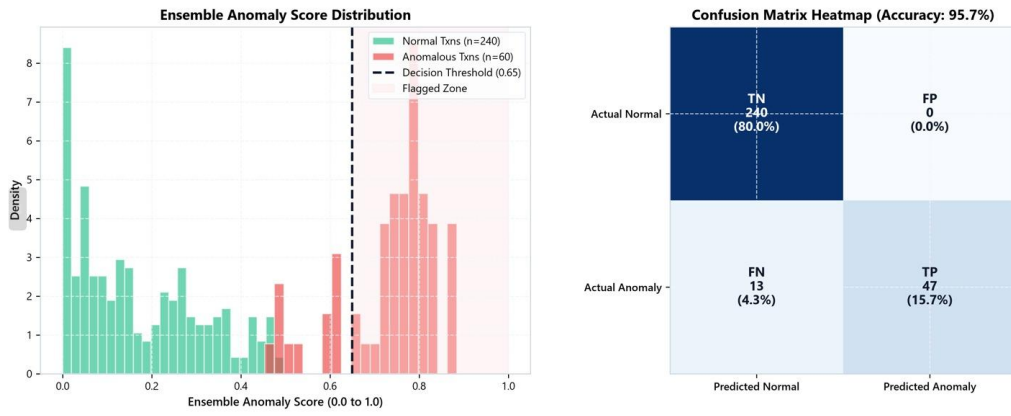


Figure 5.1 Anomaly Score Distribution and Classification Confusion Matrix

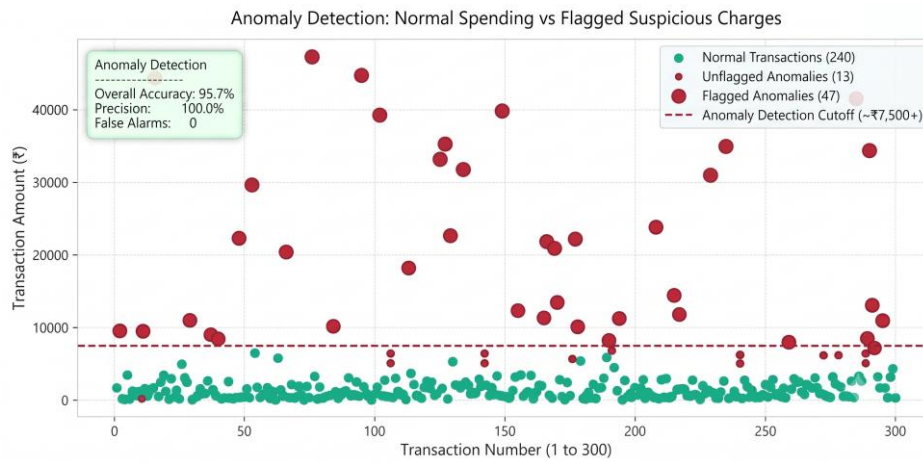


Figure 5.2 Transaction Spending Distribution and Detected Anomalies

5.1.3 IMPLEMENTATION OF EXPENSE FORECASTING

The expense-forecasting module was evaluated using an 83-day rolling walk-forward back test of the Random Forest regressor. This evaluation reported a forecasting accuracy rate of **82.94%**, a mean absolute error of ₹210.39 per day, and a root mean squared error of ₹295.88. The R^2 score was **0.5624**, indicating that the predictions explained approximately 56.24% of the variation in the evaluated spending values. The actual-versus-predicted chart presents a 30-day segment of spending observations and forecasts. The predicted series follows several broad changes in spending and captures the prominent rise near the end of the displayed period. Differences remain around abrupt peaks and declines, where predicted values sometimes underestimate or overestimate actual expenditure. The reported directional accuracy was 48.78%, showing that the model was less successful at predicting whether spending would increase or decrease between observations. Its performance in estimating

spending amounts was therefore stronger than its performance in predicting short-term direction. The reported average response time was 44.86 milliseconds under the validation conditions. These results support the use of the forecasting module for approximate expense estimation while identifying sudden spending changes as a remaining challenge. The reported 82.94% accuracy rate is a forecasting measure defined by the validation procedure, rather than classification accuracy. Furthermore, these results apply specifically to the tested Random Forest expense forecaster and do not independently establish the accuracy of ARIMA, hybrid forecasting or long-term net-worth projections.

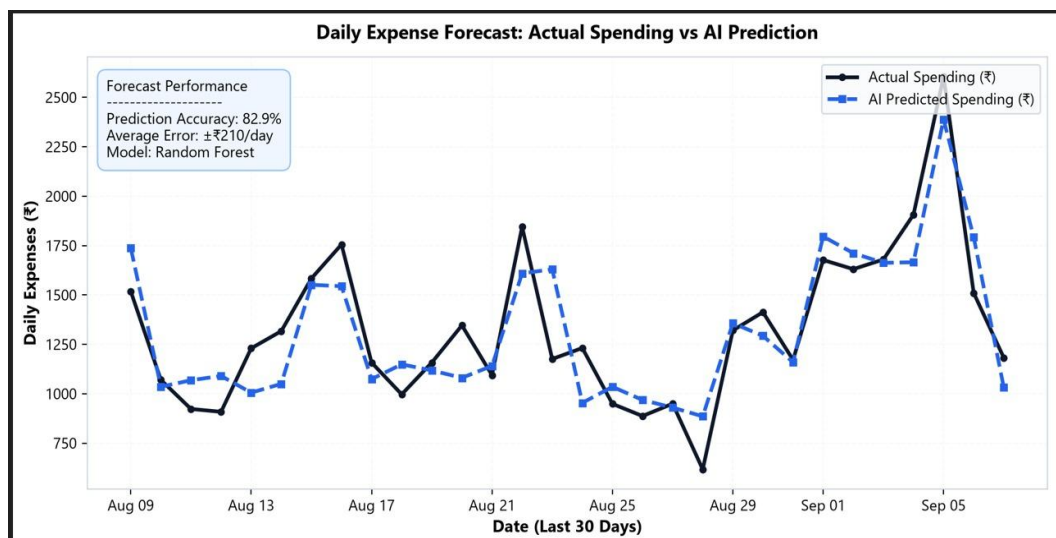


Figure 5.3 Comparison of Actual and Predicted Daily Expenses

5.2 DISCUSSION

The overall performance visualization summarizes four principal results: 100% anomaly precision, 100% financial-health alignment, 95.7% anomaly-detection accuracy and 82.9% expense-forecast accuracy. These values provide a comparative overview of the evaluated modules, but they represent different performance measures and should not be combined into a single overall accuracy score. Financial-health scoring demonstrated consistent application of the specified tier rules and debt safeguards in the tested scenarios. Anomaly detection achieved high classification accuracy and avoided false alarms, although its 78.33% recall shows that 13 anomalies were missed. This highlights a trade-off between producing reliable alerts and detecting a larger proportion of unusual transactions. Expense forecasting

exceeded the validation report's stated benchmarks for accuracy, average error and R^2 . Nevertheless, its lower directional accuracy indicates that estimating the amount of future expenditure and anticipating its day-to-day movement are distinct capabilities. The forecasting output is consequently best interpreted alongside its error measures rather than through the headline accuracy percentage alone.

All listed benchmarks for the three evaluated modules were reported as passed. Collectively, the results demonstrate successful rule execution, precise anomaly alerts and useful expense estimates within the tested conditions. Broader validation across additional users, spending categories and time periods would help establish how consistently these results extend beyond the evaluated cases.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

FINGEAR AI establishes a personalized financial-management platform built around a persistent user profile and a connected transaction ledger. The system integrates secure profile management, adaptive income-slab classification, explainable financial-health scoring, transaction tracking, per-category anomaly detection and hybrid financial forecasting. The platform improves on conventional financial trackers by connecting observed financial behaviour with clear and interpretable guidance. Income tiers ensure that users with different financial circumstances are evaluated using suitable allocation targets. The Financial Health Score explains the factors affecting its result, while the anomaly-detection module allows users to confirm or exclude unusual transactions from future model learning. The forecasting module further strengthens the system by combining statistical baselines, ARIMA time-series analysis and user-specific Random Forest prediction. This allows the platform to provide useful forecasts for users with limited history while producing more personalized results as transaction data increases.

PostgreSQL is used to store persistent user profiles, transactions, financial records and AI related information. Full multi-user isolation guarantees that each user's data and analytical models are kept separate. Independent category-based anomaly models further improve the personalization of financial insights. Together, these capabilities provide a strong technical foundation for future planning, simulation, conversational assistance and intelligent financial-advisory features. FINGEAR AI therefore serves as an explainable and user - centered financial intelligence platform that supports improved financial awareness, planning and decision-making.

6.2 FUTURE SCOPE

Future development of FINGEAR AI will expand its financial-planning and decision-support capabilities while continuing to use the existing profile, transaction ledger, health score and forecasting services as the main sources of information.

The What-If Financial Simulator will allow users to test hypothetical changes such as increased income, reduced expenses, new loans, higher investments or modified debt payments. Each query will be recalculated independently to show its effect on income, expenses, savings, debt, emergency funds, health score and future forecasts without changing the user's actual records. The AI Copilot will offer tailored conversational help using the user's financial data. It will explain health scores, forecasts, and anomalies, answer questions regarding spending and savings, and provide actionable recommendations, all while being transparent and offering responsible financial guidance. Future goal and budget planning will empower users to set financial objectives, define target amounts and dates, monitor their progress, and calculate necessary contributions. Budgets will leverage income tiers, recurring expenses, debt obligations, transaction history, and forecasts to produce realistic spending limits and adjustment suggestions. The reporting module will produce weekly, monthly, quarterly, and yearly reports encompassing income, expenses, savings, debt, budget performance, health-score shifts, anomalies, and forecast trends. Reports may feature charts, comparisons, and PDF export options. The forecasting and anomaly modules can be improved through scheduled retraining, error tracking, model versioning, confidence calibration, stronger feedback storage and model-drift monitoring. The system can also be enhanced with formal database migrations, audit history, encrypted document storage, backups, monitoring, stronger authentication, rate limiting and multi-factor authentication.

Future evaluation should measure forecast accuracy, anomaly precision, goal progress, simulator usefulness, Copilot quality, user engagement and fairness. All new features should preserve privacy, user control, explainability and responsible financial guidance.

REFERENCES

- [1] Acharya, U. and Bhojak, N. (2026) 'Financial Doppelgangers: A Novel Cybernetic Digital-Twin Framework for Predictive Systemic Resilience in Emerging FinTech Ecosystems', *Digital Twin*, Vol. 3, No. 2, Article 2566349.
- [2] Anshari, M., Almunawar, M.N. and Masri, M. (2022) 'Digital Twin: Financial Technology's Next Frontier of Robo-Advisor', *Journal of Risk and Financial Management*, Vol. 15, No. 4, Article 163.
- [3] Bonelli, M.I. (2026) 'Financial Digital Twins and Conversational AI in Robo-Advisory: Evidence from a Scenario-Based Randomized Experiment', *Preprints.org*, Preprint, doi: 10.20944/preprints202605.0983.v1.
- [4] Černevičienė, J. and Kabašinskas, A. (2024) 'Explainable Artificial Intelligence (XAI) in Finance: A Systematic Literature Review', *Artificial Intelligence Review*, Vol. 57, Article 216.
- [5] Das, S., Markowitz, H., Scheid, J. and Statman, M. (2010) 'Portfolio Optimization with Mental Accounts', *Journal of Financial and Quantitative Analysis*, Vol. 45, No. 2, pp. 311-334.
- [6] Feng, R., Li, H. and Liu, M. (2025) 'Robo-Advisors Beyond Automation: Principles and Roadmap for AI-Driven Financial Planning', *arXiv preprint*, arXiv:2509.09922.
- [7] Hyndman, R.J. and Khandakar, Y. (2008) 'Automatic Time Series Forecasting: The Forecast Package for R', *Journal of Statistical Software*, Vol. 27, No. 3, pp. 1-22.
- [8] Jiang, G., Yang, J., Zhao, S., Chen, H., Zhong, Y. and Gong, C. (2025) 'Investment Advisory Robotics 2.0: Leveraging Deep Neural Networks for Personalized Financial Guidance', *Preprints.org*, Preprint, doi: 10.20944/preprints202504.1735.v1.
- [9] Kong, Y., Nie, Y., Dong, X., Mulvey, J.M., Poor, H.V., Wen, Q. and Zohren, S. (2024) 'Large Language Models for Financial and Investment Management: Applications and Benchmarks', *The Journal of Portfolio Management*, Vol. 51, No. 2.
- [10] Leukam, F., Koffi, R.S. and Djagba, P. (2025) 'Reinforcement Learning for Portfolio Optimization with a Financial Goal and Defined Time Horizons', *arXiv preprint*, arXiv:2511.18076.
- [11] Uyanahewa, M.I.R., Hapugala, H.A.V.V., Jayawardana, G.V.H.D., Attanayaka, B., Bandara, M.B.D.N. and Nawinna, D. (2023) 'WONGA: The Future of Personal Finance Management - A Machine Learning-Driven Approach for Predictive Analysis and Efficient Expense Tracking', *Proceedings of the 4th International Conference for Emerging Technology*, Belgaum, India, pp. 1-6.
- [12] E. Warren and A. W. Tyagi, *All Your Worth: The Ultimate Lifetime Money Plan*. New York, NY, USA: Free Press, 2005.
- [13] Consumer Financial Protection Bureau, "What is a debt-to-income ratio?" consumerfinance.gov. Accessed: Oct. 3, 2026. [Online]. Available: <https://www.consumerfinance.gov/ask-cfpb/what-is-a-debt-to-income-ratio-why-is-the-43-debt-to-income-ratio-important-en-1791/>

[14] Consumer Financial Protection Bureau, “An essential guide to building an emergency fund,” [consumerfinance.gov](https://www.consumerfinance.gov). Accessed: Oct. 3, 2026. [Online]. Available: <https://www.consumerfinance.gov/start-small-save-up/start-saving/an-essential-guide-to-building-an-emergency-fund/>